EFY TESTED
S.C. Dwivedi

Basic Installation Of **Access Control System**

S. Aravind

This article describes the basic installation of an RFID and password-based access control system. Access control systems allow only the authorised persons to enter or exit the premises. An access control system may include locks, turnstiles, biometric readers, face recognition readers, RFID readers, boom barriers, etc.

The materials required for this project include:

- Magnetic lock with a mounting bracket (fail-safe type lock)
- DC power supply adaptor or switched-mode power supply (SMPS) (12V, 3A)
- Access control reader
- Exit push button
- Connecting wires (Cat 6 or copper wires, as required)

Circuit and working

The block diagram of the access control system is shown in Fig. 1. It is built around a magnetic lock with mounting brackets, 12V SMPS/adaptor for power supply, access control device, exit push button, RFID tags and cables. It includes 3- and 6-pin connectors along with 3- and 6-wire cables to connect the access control device. The complete wiring diagram for the project is shown in Fig. 2.

Relay connections. Let us go through the relay concept before the actual installation. The relay (refer Fig. 3) has three contacts: NO, NC and COM, which are usually referred to as normally open, normally close, and common terminal, respectively. COM usually remains connected to NC terminal. When the relay is activated, the COM terminal gets connected to NO terminal, breaking away from NC terminal. Most of the access control readers, push buttons and locks have an inbuilt relay circuit for the desired connections.

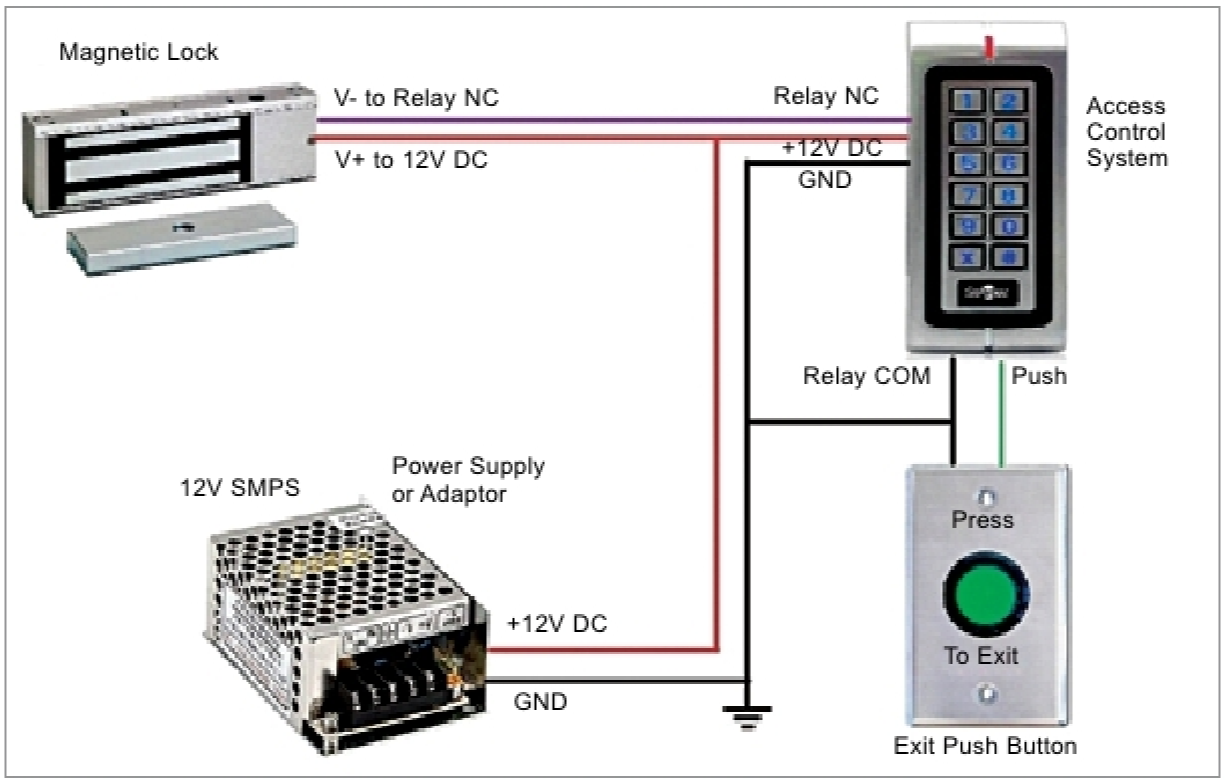

Fig. 1: Block diagram of author's prototype

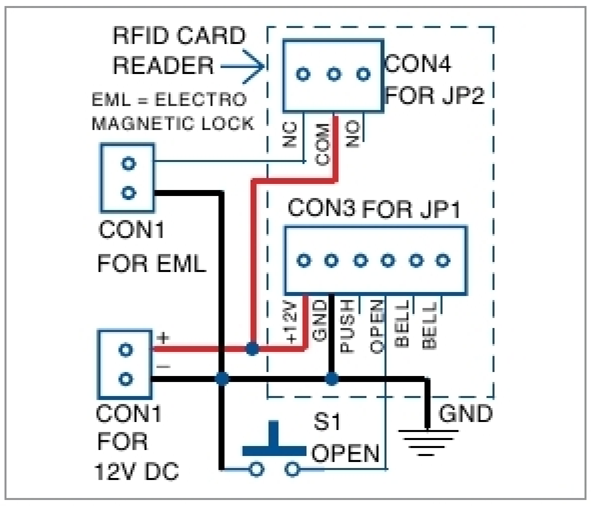

Fig. 2: Wiring diagram of access control system

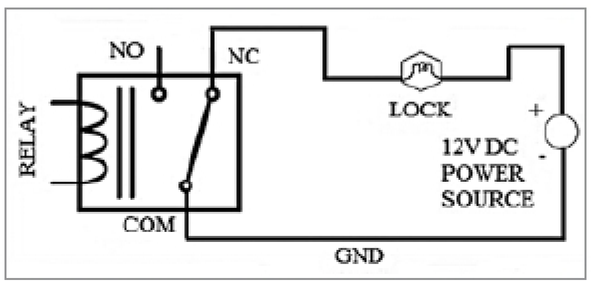

Fig. 3: Relay connections

Access control device. Access control device and magnetic lock are the main components required for the installation. The standalone access control device (model number SA32-E) used in this project is shown in Fig. 4. It is a contactless RFID smart card reader and password verification device. It enables three verification modes (RFID card only, PIN only, and card & PIN) to ensure maximum safety and flexibility. With one relay output (optional) and a push-button (low) output, one door button port, and one doorbell port, the device is simple but effective for use in residential housing, offices, and other public facilities.

When the magnetic lock is directly connected with a +12V supply and ground (GND) across the terminals, the lock functions normally (that is, the door remains locked). Note that there are two main types of magnetic locking devices: fail-safe and fail-secure. A fail-secure locking device gets locked when power is off. Fail-safe locking devices are locked when power is on and gets unlocked when power goes off. This project uses a fail-safe locking device.

The locking device must be configured in such a way that if a person uses the RFID tag (or biometric) to